

1. [6%] Please draw the diagram of setup and hold time check if a register in CLK1 domain is writing signal into a register in CLK2 domain under the set_multicycle_path timing constrain:

(1) (2 points)

set_multicycle_path 1 -end -from CLK1 -to CLK2

set_multicycle_path 0 -hold -start -from CLK1 -to CLK2

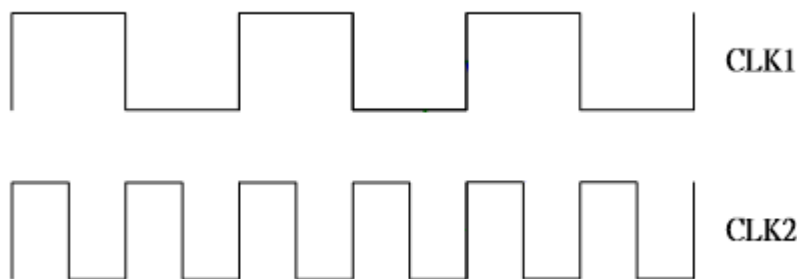
(2) (4 points)

set_multicycle_path 3 -start -from CLK1 -to CLK2

set_multicycle_path 2 -hold -start -from CLK1 -to CLK2

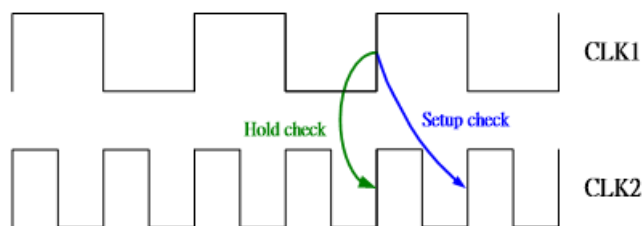
Given that the waveform relationship of clk1 and clk2 is:

(if the first rising edge of CLK1 is perfectly aligned to the first rising edge of CLK2 and the period of CLK1 is twice longer than the period of CLK2)

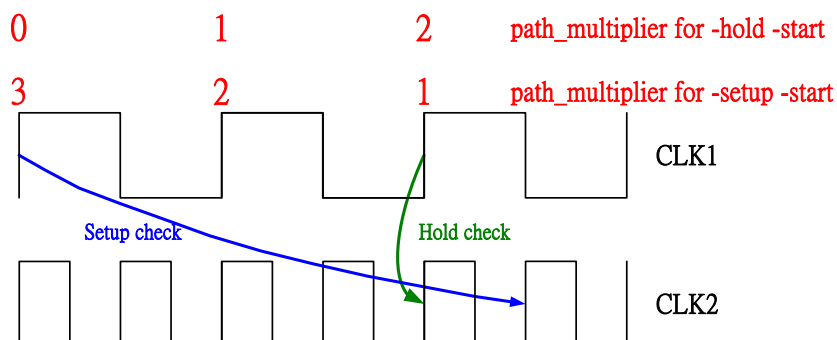


Solution:

(1)



(2)



2. [9%] Please explain the main difference between double flop synchronizer and XOR double flop synchronizer if we use them in a CDC (clock domain crossing) design and why do we need to use synchronizer in a CDC design.

(9 points)

Solution:

Main difference:

2-FF: flag signal must be high long enough so the synchronized signal won't be lost because of metastability issue. (3 points)

XOR: flag signal can be high for only one cycle because of the feedback XOR circuit, and the output signal will be high for only one cycle. (3 points)

Why do we need to use synchronizer in a CDC (clock domain crossing) design?

The unstable status due to non-ideal data transition is called **metastability**. CDC designs will inevitably face this problem, because the delay between to register of two different clk domain will alter through time. (3 points)

3. [10%] Actually, we can use report_power to get our power report in the design compiler. However, we usually get our power report by using PrimeTime. Why should we using PrimeTime to get the power report? What kind of file should we prepare if we want to get the power report by using PrimeTime?

Ans:

[4%] We can get more accurate power analysis by using PrimeTime than by using the design compiler. Because there are the value changes of the signals in the switching activity file, PrimeTime can analyze the power of the design more accurately.

[6%] Prepared files:

Phase 1 -- relative file preparation

- Technology library
- Gate-level netlist
- Switching activity file (Value Change Dump file or FSDB file)
- Parasitic file (after APR) (RSPF, SPEF, or DSPF file) optional
- Synopsys design constraint (SDC) file
- Standard delay format (SDF) file

4. [6%] Please list two problems which will happen when a signal propagates between two power domains at different voltage levels. What kind of circuits should we use to deal with these problems?

Ans:

[4%] 、 [2%]

✓ **Connecting 2 power domains at different voltage levels can cause design issues**

- Timing inaccuracy
- Signals are not propagated

✓ **A level shifter is required**

5. [20%] For (a) to (g), please fill in the blank

For (h) to (k), please answer the following question

(h) : how many packed array is declared in ALU.sv? how many bits is used to generate these packed array?

(i) : please indicate the problem of type FSM_STATE.

(j) : please indicate the problem of type STAGE.

(k) : please rewrite Piece.v so that we can put it inside the program of PATTERN.sv

a-1 SystemClock / .clk(SystemClock)

a-2 input bit clk / input clk

b inf.PATTERN

c inf.ALU

d modport

e import

f ::*

g package

h 32, 996 (1 point if your answer is 27, 840)

I Default type for enum is integer. Either remove “3’d” or declare “logic [2:0]

“ may solve this problem.

J s_g will be automatically assigned as 4, which will cause conflict with s_h and syntax error.

K replace the always block with a forever block inside an initial block.

```
1 initial begin
2     forever @(negedge clk) begin
3         if(inf.out_valid)begin
4             $display("out_valid is high");
5         end
6     end
7 end
```

6. [6%] What are code coverage and functional coverage?

Ans:

[3%] Code coverage: This will give information about how many lines are executed, how many times expressions, branches executed. This coverage is collected by the simulation tools. Users use this coverage to reach those corner cases which are not hit by the random test cases. Users have to write the directed test cases to reach the missing code coverage areas.

[3%] Functional coverage: This coverage will be defined by the user. User will define the coverage points for the functions to be covered in DUT. This is completely under user control.

7. [9%] If we want to check whether our design hits the specification, we can write the assertion in the check file. There are two signals, A and B, in the design. Please finish three assertions as the below.

- (1) A happens then 2 or more cycles later B happens
- (2) If A happens, then B must happen at the same cycle.
- (3) If A happens, then B must happen at the next cycle.

Assertion1: assertion property (@(posedge clk)) ;

Assertion2: assertion property (@(posedge clk)) ;

Assertion3: assertion property (@(posedge clk)) ;

Ans:

[3%] Assertion1: assertion property (@(posedge clk)) A ##[2:\$] B;

[3%] Assertion2: assertion property (@(posedge clk)) A -> B;

[3%] Assertion3: assertion property (@(posedge clk)) A => B;

8. [8%] Why do we need to add pad filler and filler cell in APR flow? Please explain the reason respectively.

A:

Pad filler: make no space between pads

Filler: Provide decoupling capacitances to complete connections in the standard cell rows

9. [8%] Please explain what is crosstalk, and list at least three solutions.

A:

Crosstalk is any phenomenon by which a signal transmitted on one circuit or channel of a transmission system creates an undesired effect in another circuit or channel.

Insert buffer in lines

Upsize driver

Congestion optimization

Limit length of parallel nets

Wider routing grid

Shield special nets

10. [4%] Briefly explain what are (a) power analysis [2%] and (b) rail analysis [2%]

Ans:

Power analysis: Analyze the power consumption of your design

Rail analysis: Analyze if the powerplan is enough to provide whole chip

11. (a) IR drop can cause the chip to fail due to (at least 3)? [4%]

(b) What can we do to solve IR drop? [4%]

- IR drop can cause the chip to fail due to
 - ◆ Performance (circuit running slower than specification)
 - ◆ Functionality problem (setup or hold violations)
 - ◆ Unreliable operation (less noise margin)
 - ◆ Power consumption(internal power)
- Prevention: adding stripes to avoid IR drop on cell's power line

Bonus Formal Verification (10%)

1. True or False: (7%)

1. (X) Formal Verification could totally replace simulation-based verification in current IC Design flow.
2. (X) We normally write our formal verification syntax in our Pattern.sv to setup our verification environment.
3. (X) Formal Verification is a verification method that could be applied only in design written in system Verilog
4. (O) Generally, we use cover , assert , and assume to setup our verification environment.
5. (O) Assume properties tell formal tool what kind of stimulus is legal, so only the matched stimulus will be considered to verify our design.
6. (O) Glue Logic is an alternative method to make our formal properties (such as assume, assert or cover) simple and easy to read.
7. (O) In code coverage, we could comment out few of some assume properties to check whether our verification environment is over-constrained or not.

Hint: code coverage is a verification methodology to test whether each statement/branch/expression has been covered.

2. Fill in the Blanks: (3%)

In Verification IP usage, there are 3 kind of Verification-Oriented configuration to set up our AXI-verification environments. (Master Mode, Slave Mode and Monitor Mode)

